Urban Air Pollution Challenge

Machine Learning Project: Predicting Daily PM2.5 Concentrations Using Machine Learning

Integrating Ground-Based Sensors, GFS (Global Forecast System) Weather Data and Sentinel-5P Satellite Observations

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Goal

The objective of this challenge is to predict PM2.5 particulate matter concentration in the air every day for each city: predict daily PM2.5 concentration for each city.
PM2.5 = fine particulate matter <2.5 µm.

Target variable: **target = daily mean** PM2.5 concentration.

Train set contains observed PM2.5 measurements.
Test set contains locations/dates for which PM2.5 must **be predicted**.

Question

Can weather and satellite-derived atmospheric variables accurately predict daily PM2.5 concentrations?

Data Preparation & Predictors (1/1)

Raw data (TrainData=82 feature and 30,557 observations, TestData=77 feature and 16136 observations)

⬇

Identify target

⬇

Remove target-derived variables

⬇

Remove ID/date variables

⬇

Handle missing values

⬇

Feature Engineering Highly correlated feature pairs (|correlation| >= 0.95): (one feature of the pair has been removed

⬇

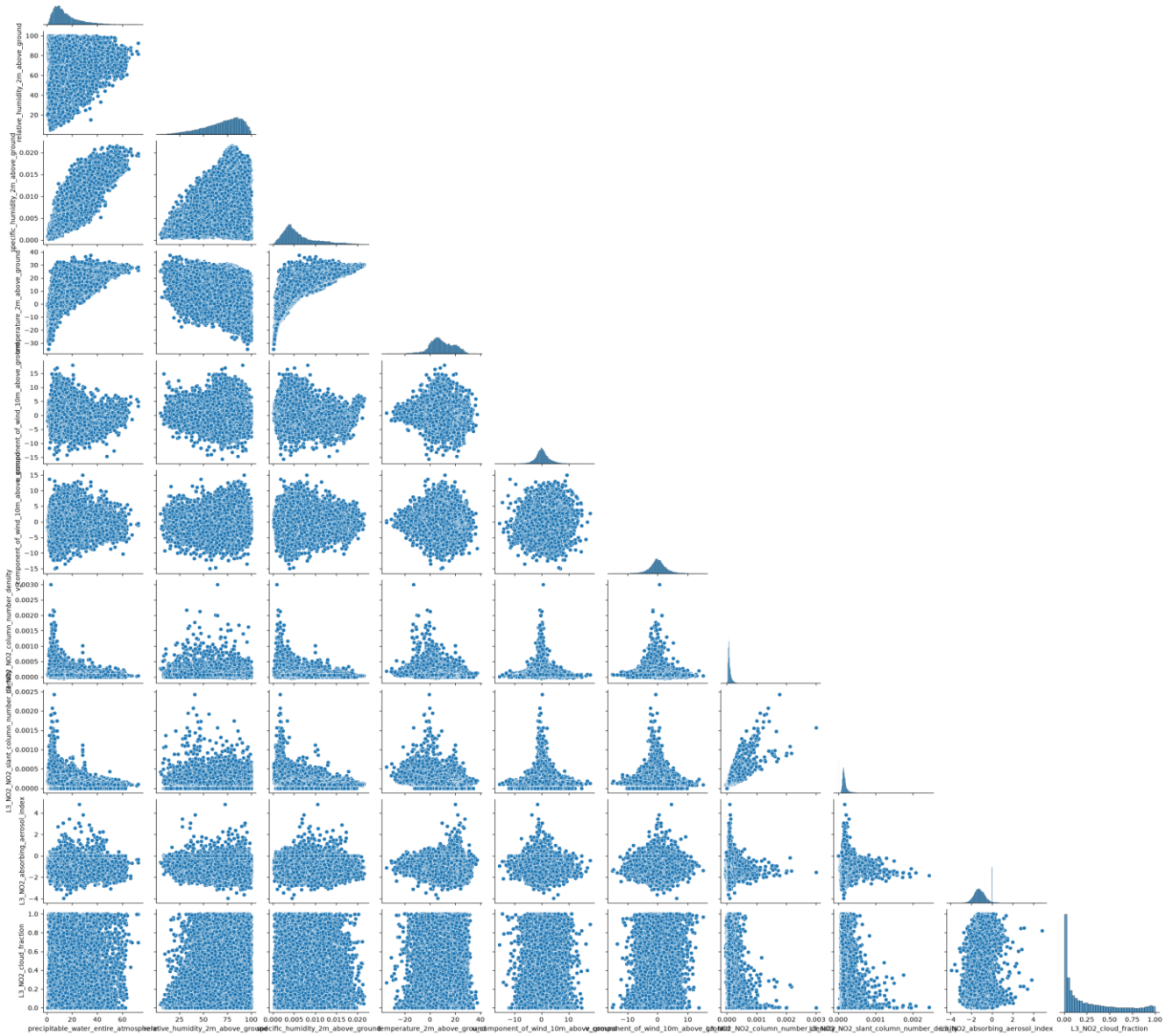
54 predictors (features)

Sources	Examples	Role
Ground sensors	PM2.5 daily mean	Target
GFS weather	Humidity, temperature, wind	Predictors
Sentinel-5P	NO_2, CO, SO_2, O_3, HCHO, CH_4, aerosols	Predictors

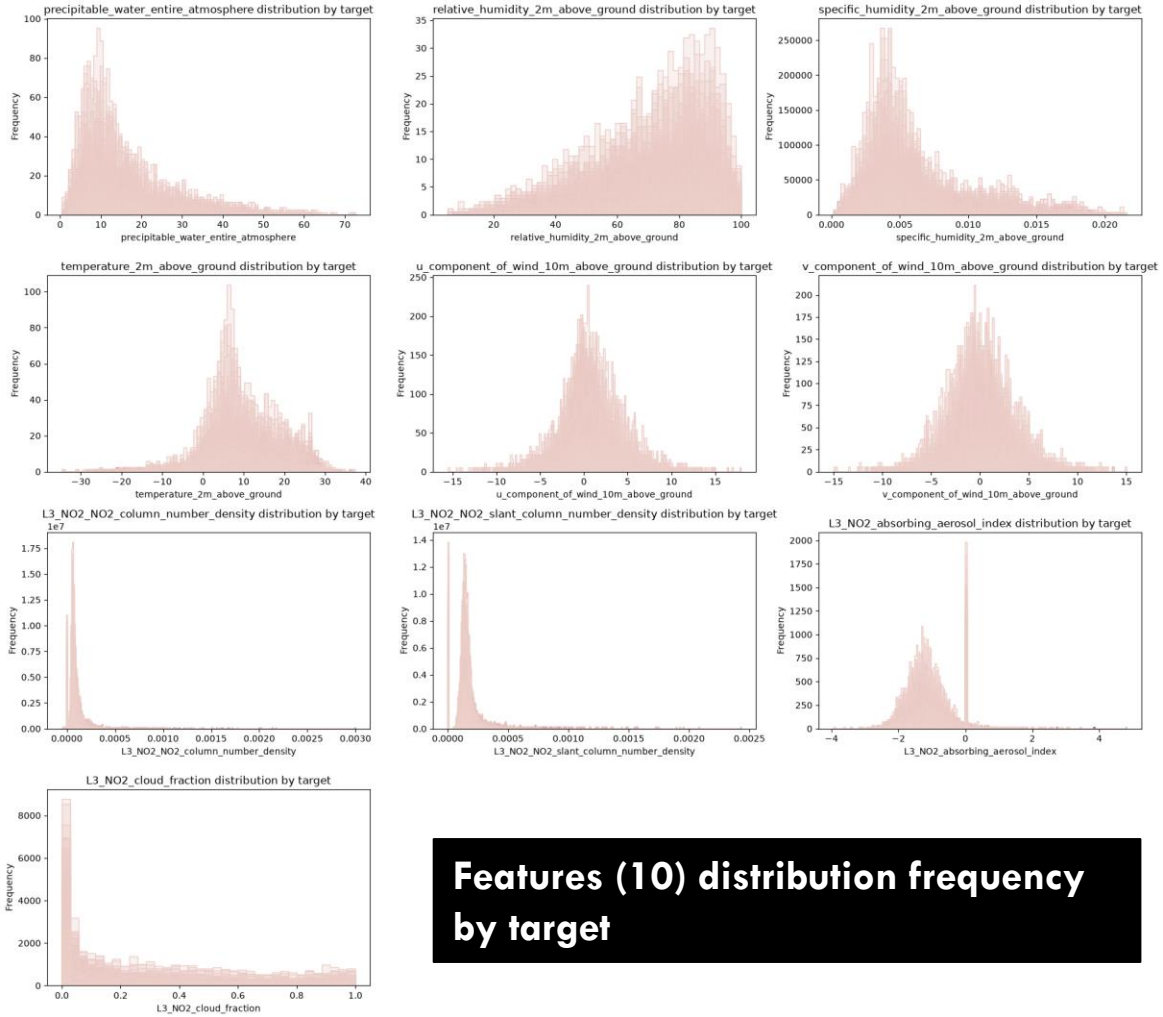
- Target (Target mean predicted PM2.5 particulate matter)
- target_min
- target_max
- target_variance
- target_count
- Place_ID
- Place_ID X Date
- Date and correlated features (20)

EDA(1/1)

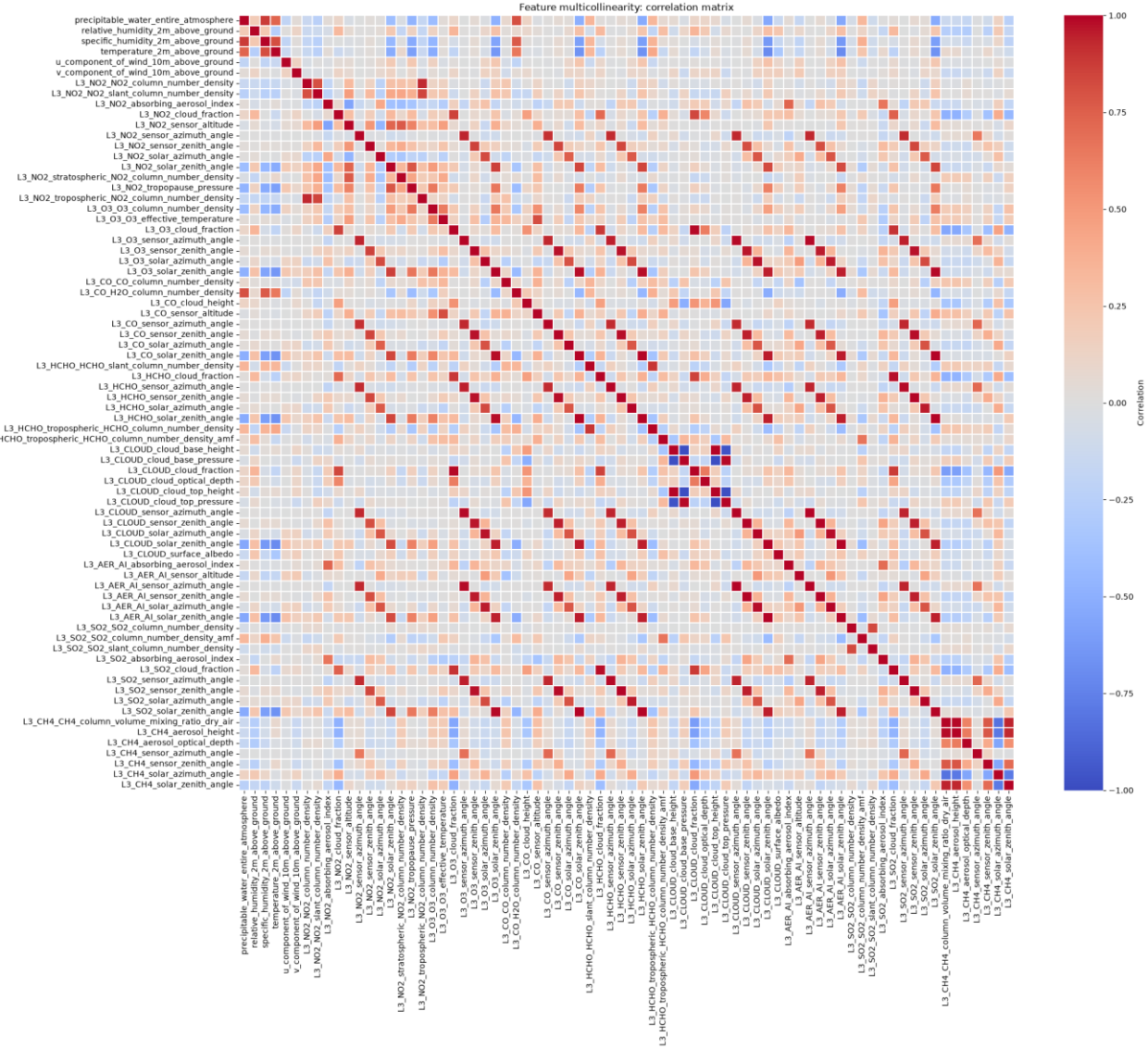
Scatterplot Matrix: show the distribution of a single variable



EDA(1/2)



Features (10) distribution frequency by target



Multicollinearity of the features

Key Insights: Prediction(1/1)

Engineering

Train / Validation Split

Preprocessing/build the pipelines

Feature Selection

Model Training

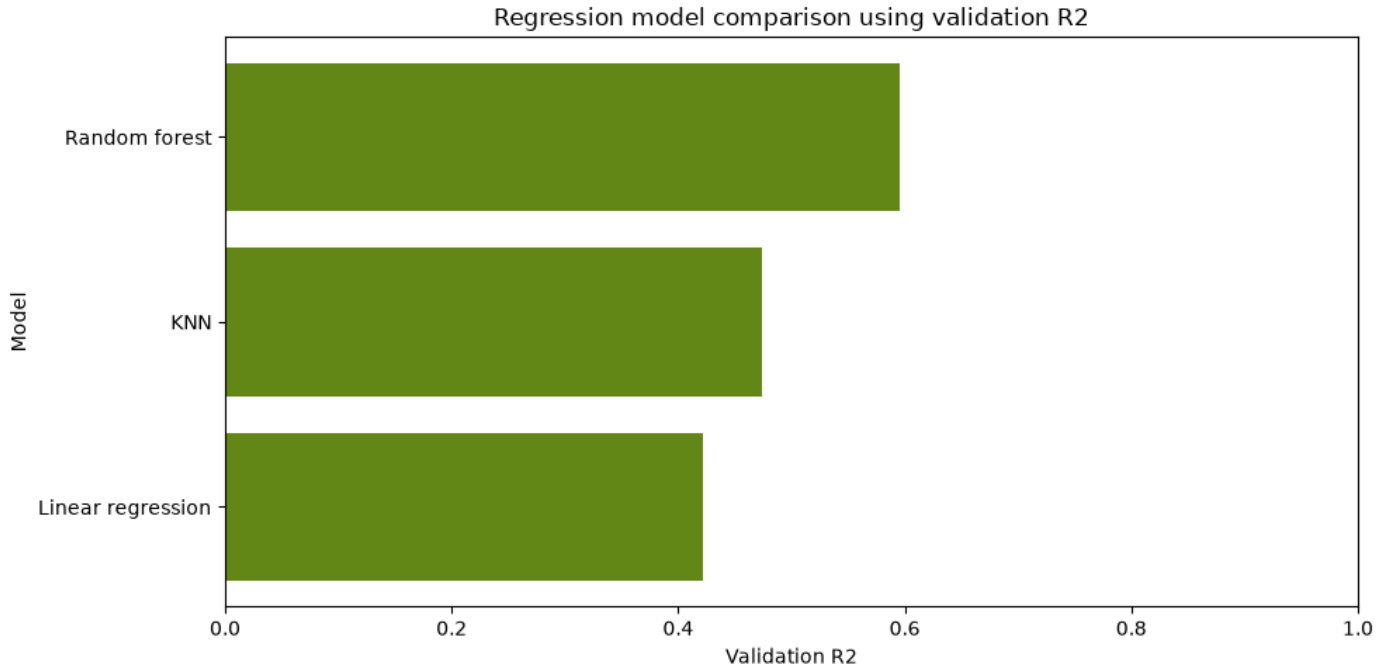
Hyperparameter / Feature Selection

Validation

Model Comparison

Random Forest, K-Nearest Neighbors (KNN), Linear Regression

Model	RMSE	MAE	R^2
Random Forest	**29.482**	**20.491**	**0.596**
KNN	33.628	23.566	0.474
Linear Regression	35.257	25.961	0.422



Key Insights: Feature Selection & Permutation importance (1/2)

54 predictors

↓

SelectKBest + GridSearchCV

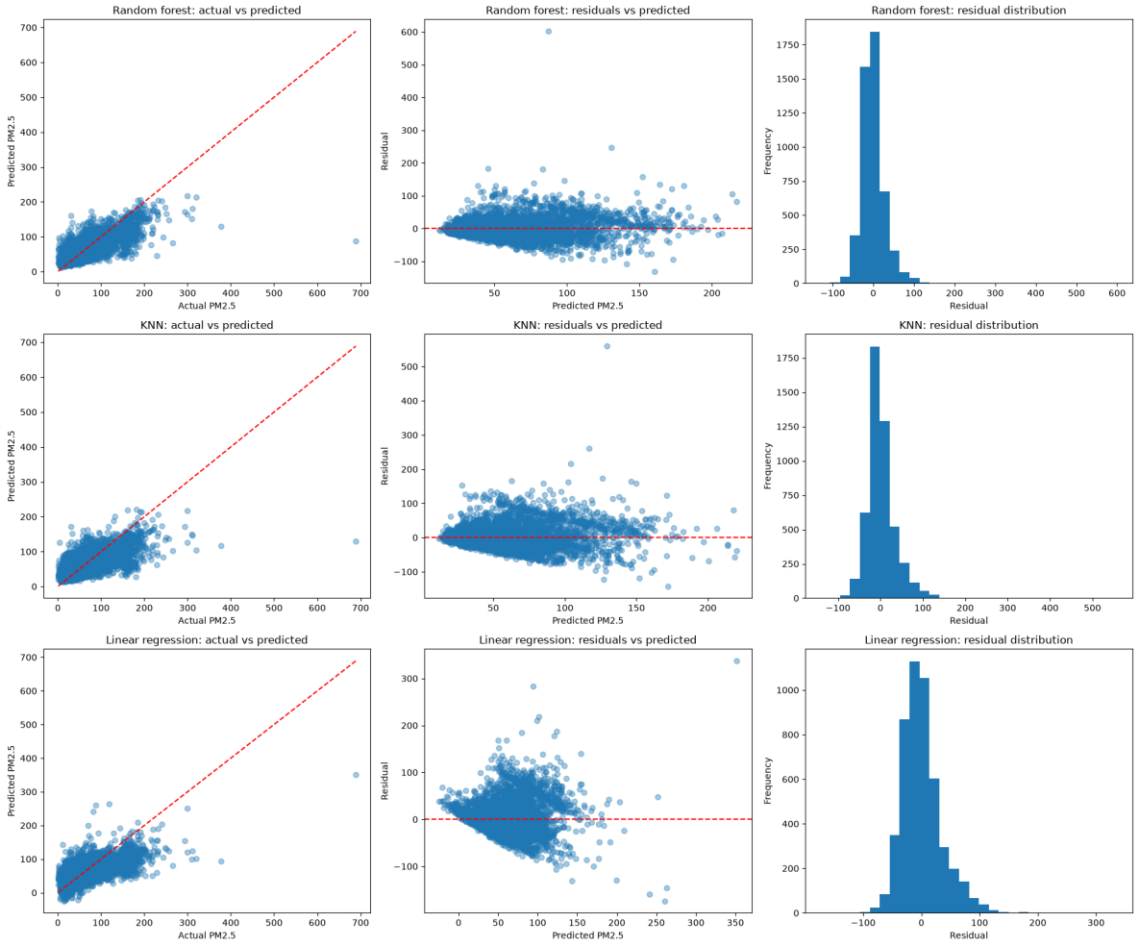
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Optimal feature subset

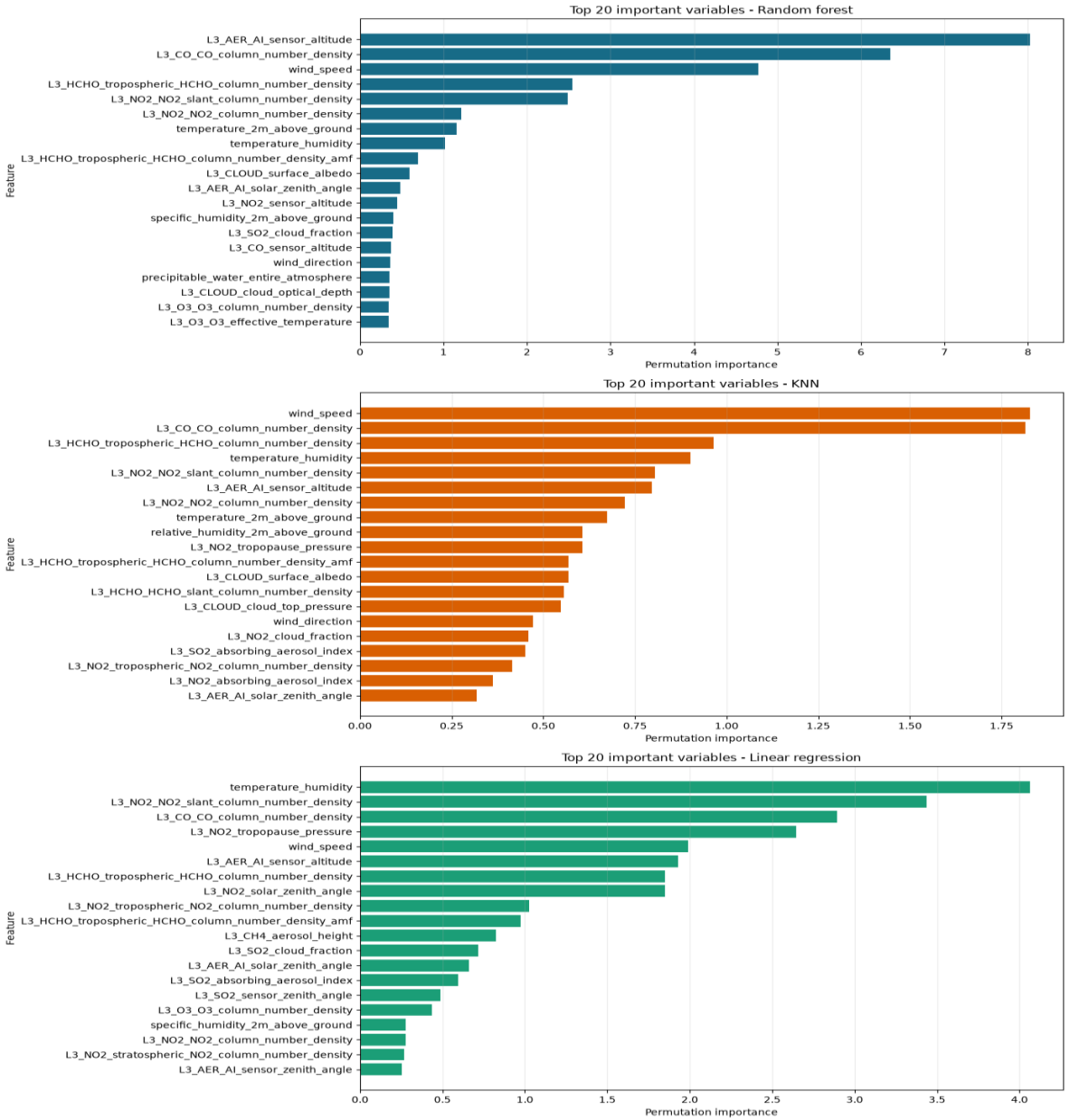
Model	Best CV RMSE	Optimal number of features (K)
Random Forest	**31.970**	**40**
KNN	34.630	40
Linear Regression	36.809	54

Model	Validation RMSE	Validation MAE	Validation R^2
Feature-selected Random Forest	**29.732**	**20.657**	**0.589**
Feature-selected KNN	32.334	22.452	**0.514**
Feature-selected Linear Regression	35.257	25.961	0.422

Key Insights: Permutation importance & Residual analysis (1/3)



Residual analysis for the models



top 20 important variables for each model

Conclusions

1. Random Forest performed best
$R^2 \approx$ **0.60** on the validation data.

2. Environmental predictors contain substantial information about PM2.5
Weather and atmospheric composition variables provide useful predictive signals.

3. Feature selection reduced dimensionality
54 predictors: **40 selected predictors for Random Forest**.

4. Feature selection did not increase predictive accuracy
Performance remained very similar, suggesting that the additional variables contained useful complementary information.

5. Final objective
The selected **Random Forest** was used to generate PM2.5 predictions for the unseen **test observations**.




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YOU !


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